NPDE Note

1 Definitions, Classification of PDEs

1.1 Big O, Classification

Big $\mathcal{O}$: $f(t)=\mathcal{O}(g(t))$ as $t\to t_0$ if $\exists\varepsilon>0, C>0$ s.t. $\forall |t-t_0|<\varepsilon, |f(t)|\le C|g(t)|$

$\Leftrightarrow \lim_{t\to t_0}[f(t)/g(t)]$ bdd ps: If not say $t\to t_0$, assume $t\to 0$

More on Big $\mathcal{O}$: **a.** $F(x)=G(x)+\mathcal{O}(H(x))$ if $F(x)-G(x)=\mathcal{O}(H(x))$

b. $F(x)=\mathcal{O}(G(x))+\mathcal{O}(H(x))$ if $F(x)=\mathcal{O}(\max\{|G(x)|,|H(x)|\})$

c. $F(x,y)=\mathcal{O}(x^m)+\mathcal{O}(y^n)$ if [Given $G(x)=\mathcal{O}(x^{\frac{m}{n}}) \implies F(x,G(x))=\mathcal{O}(x^m)$], 且另一项同理

Classification: For $au_{xx}+bu_{xy}+cu_{yy}+du_x+eu_y+fu+g=0$ $(a,b,c)\neq(0,0,0)$

1. **Hyperbolic:** $b^2-4ac>0$ Admits non-trivial plane *wave solutions* (Propagation)
2. **Parabolic:** $b^2-4ac=0$ **and** we can get $2cd\neq be$ or $2ae\neq bd$
3. **Elliptic:** $b^2-4ac<0$ *No* non-trivial plane *wave solutions* (No Propagation)

Remark: 对变量 x,y 进行缩放后不会改变 PDE 类型

1.2 Norm

Weighted 2-Norm: $\|\mathbf{u}\|_{2,\Delta x}=\sqrt{(\Delta x)^d}\|u\|_2$ where d s.t. $\mathbf{u}:\mathbb{R}^d\to\mathbb{R}^n$

- For $d=1$ and $\Delta x=\frac{D}{N}$, then: $\forall i, u_i=\mathcal{O}((\Delta x)^p)\Rightarrow \|\mathbf{u}\|_{2,\Delta x}=\mathcal{O}((\Delta x)^p)$
- For A symmetric: $\|A^{-1}\mathbf{w}\|_2\le\frac{1}{|\lambda_1|}\|\mathbf{w}\|_2$ $0<|\lambda_1|\le\cdots\le|\lambda_n|$

Property: If $\|\mathbf{u}\|_2\le\|\mathbf{w}\|_2$, then $\|\mathbf{u}\|_{2,\Delta x}\le\|\mathbf{w}\|_{2,\Delta x}$

New Norm: The norm $\|\mathbf{u}\|:=\max_n\|\mathbf{u}^n\|_{2,\Delta x}$ where $\mathbf{u}^n=[u_1^n,u_2^n,\dots,u_{M-1}^n]^\top$

Which means: Infinity norm in time, weighted 2-norm in space. 主要用于双变量 PDE

- For $\tau_m^n=\mathcal{O}((\Delta x)^a)+\mathcal{O}((\Delta t)^b)$, $\Delta x=\frac{b-a}{M}$, $\Delta t=\frac{T}{N}$, assume $\Delta t=K(\Delta x)^\beta$, then:

$\max_n\|\underline{\tau}^n\|_{2,\Delta x}=\mathcal{O}((\Delta x)^{\min\{a,b\beta\}})$

1.3 Linear Algebra

Definite: **Positive Definite:** $\lambda_i>0,\forall i$ **Negative Definite:** $\lambda_i<0,\forall i$

Symmetric: If A **symmetric**, then $A=Q\Lambda Q^\top$ where Q orthogonal, Λ diagonal with eigenvalues.

Normal: Real matrix A is **normal** if $AA^\top=A^\top A$ The following are properties of it:

1. A is orthogonally diagonalizable, i.e. $A=Q\Lambda Q^\top$ where Q orthogonal, Λ diagonal with eigenvalues.
2. Exists an eigenvectors of A: $\{\mathbf{v}_i\}$ s.t. $\mathbf{v}_i^*\mathbf{v}_j=\delta_{ij}$ and Symmetric $\Rightarrow$ Normal

2 Global/Local Error, Convergence, Consistency, Stability

Exact & Approx. Sol: U_n: **approx. n-th time-step sol**; u_n: **exact sol at n-th time-step**.

Global Error $(e_n,\mathbf{e}(\Delta t))$: $e_n=U_n-u(t=n\Delta t)$ $\mathbf{e}(\Delta t)=[e_0,e_1,\dots,e_N]^\top$

Convergence: A numerical method is convergent if $\lim_{\Delta t\to 0}\|\mathbf{e}(\Delta t)\|=0$

If $\lim_{\Delta t\to0}\frac{\|\mathbf{e}(\Delta t)\|}{(\Delta t)^p}=C$ (i.e. $\|\mathbf{e}(\Delta t)\|=\mathcal{O}((\Delta t)^p)$) $\Rightarrow$ **p-th order accurate**

Local Truncation Error (τ_n): 数值分析的离散迭代公式, 将 U_n 替换为 u_n 后的误差 *(LHS - RHS)*.

- e.g. For $\frac{U_{n+1}-U_n}{\Delta t}=f(U_n)$, $\tau_{n+1}=\frac{u_{n+1}-u_n}{\Delta t}-f(u_n)$ $\tau_0=0$
- e.g. 我们可以把线性数值方法写成 $Au=b$, A matrix, $u=[U_1,\dots,U_{N-1}]^\top$.

Then: $\tau=Au_{\text{exact}}-b$ where $u_{\text{exact}}=[u_1,\dots,u_{N-1}]^\top$, $\tau=[\tau_1,\dots,\tau_{N-1}]^\top$

ps: $Ae=-\tau$ **where** $e=u-u_{\text{exact}}$ Define $\tau(\Delta t)=[\tau_1,\dots,\tau_{N-1}]^\top$

Consistency: If $\lim_{\Delta t\to0}\|\tau(\Delta t)\|=0$, the method is consistent.

Linear Equation: Goal: Finally change ODE/PDE into $Au=b$ where **A is matrix**, $u=[U_1,\dots,U_{N-1}]^\top$

Stability: A numerical method is stable if $\exists\Delta x_0$ s.t. For the Linear System $Au=b$:

1. $A(\Delta x)$ is invertible $\forall\Delta x<\Delta x_0$
2. $\|[A(\Delta x)]^{-1}\mathbf{w}\|\le C\|\mathbf{w}\|$ for all $\mathbf{w}\in\mathbb{R}^{N-1}$

Property: Consistent + Stable $\Rightarrow$ Convergent (same norm!) and $\|\mathbf{e}(\Delta t)\|=\mathcal{O}(\|\tau(\Delta t)\|)$

3 Intro & Method of Lines

Taylor Series: $f(t+\Delta t)=f(t)+f'(t)\Delta t+\frac12 f''(t)(\Delta t)^2+\cdots+\mathcal{O}((\Delta t)^N)$ **ps:** $u_{n+1}=u(n\Delta t+\Delta t)$

$$f(t)=f(a)+f'(a)(t-a)+\frac{f''(a)}{2!}(t-a)^2+\cdots+\frac{f^{(N-1)}(a)}{(N-1)!}(t-a)^N+\mathcal{O}((t-a)^N)$$

Euler Methods: Forward: $\frac{du}{dt}|_{n\Delta t}\approx\frac{U_{n+1}-U_n}{\Delta t}$; Backward: $\frac{du}{dt}|_{n\Delta t}\approx\frac{U_n-U_{n-1}}{\Delta t}$

Leapfrog: $\frac{du}{dt}|_{n\Delta t}\approx\frac{U_{n+1}-U_{n-1}}{2\Delta t}$ **Trapezoidal:** $\frac12[\frac{du}{dt}|_{n\Delta t}+\frac{du}{dt}|_{(n+1)\Delta t}]\approx\frac{U_{n+1}-U_n}{\Delta t}$

Centred Difference: $\frac{d^2u}{dt^2}|_{n\Delta t}\approx\frac{U_{n-1}-2U_n+U_{n+1}}{(\Delta t)^2}$

Method of Lines: Using $U_m(t)\approx u(a+m\Delta x,t)$, $U_m^n\approx u(a+m\Delta x,n\Delta t)$ a 为空间起点, 不重要

1. **For 1D: a.** 先对空间 x 离散化, 使用 U_m 表达, 此时 $\frac{d\mathbf{U}}{dt}=\dots$ where $\mathbf{U}=[U_1(t),U_2(t),\dots,U_{M-1}(t)]^\top$

   b. 对时间 t 使用 ODE 离散化, 使用 U_m^n 表达, 得到 Linear Equation.
2. Example: Crank-Nicolson for Heat Eq. $u_t=\kappa u_{xx}$: 对 x 用 centred difference; 对 t 用 trapezoidal:

$$\frac{U_m^{n+1}-U_m^n}{\Delta t}=\frac{\kappa}{2}\left(\frac{U_{m+1}^{n+1}-2U_m^{n+1}+U_{m-1}^{n+1}}{(\Delta x)^2}+\frac{U_{m+1}^n-2U_m^n+U_{m-1}^n}{(\Delta x)^2}\right)$$

$$\Rightarrow -rU_{m-1}^{n+1}+(2+2r)U_m^{n+1}-rU_{m+1}^{n+1}=rU_{m-1}^n+(2-2r)U_m^n+rU_{m+1}^n,\quad r=\frac{\kappa\Delta t}{(\Delta x)^2}$$

Solve it: By iteration: e.g. $A\mathbf{U}^{n+1}=\mathbf{B}\mathbf{U}^n$, where $\mathbf{U}^n=[U_1^n,U_2^n,\dots,U_{M-1}^n]^\top$.

Truncation for MoL: 对于 τ_m, 离散 x 后 U_m 替换为 u_m; 对于 τ_m^n, 再离散 t 后 U_m^n 替换为 u_m^n.

e.g. For Crank-Nicolson above: $\tau_m=\frac{\partial u_m}{\partial t}-\kappa\frac{u_{m+1}-2u_m+u_{m-1}}{(\Delta x)^2}$

e.g. For Crank-Nicolson above: $\tau_m^n=\frac{u_m^{n+1}-u_m^n}{\Delta t}-\frac{\kappa}{2}(\cdots)$

4 Lax-Richtmyer Stability, von Neumann Analysis

Abs. Stability for ODEs: For ODEs $\frac{du}{dt}=A\mathbf{u}$, $\mathbf{u}$ vector. A 1-step method for t (i.e. $\mathbf{U}^{n+1}=B\mathbf{U}^n$)

It's absolutely stable if $\|B\mathbf{w}\|_2\le\|\mathbf{w}\|_2$ for all $\mathbf{w}\in\mathbb{R}^{N-1}$

Property: If B normal, then absolutely stable $\Leftrightarrow|\lambda_i(B)|\le1,\forall i$

Property: Absolute stability $\Rightarrow$ Lax-Richtmyer stability in 2-norm and weighted 2-norm

Lax-Richtmyer Stability: For PDE which discrete form $\frac{1}{\Delta t}\mathbf{U}^{n+1}=\frac{1}{\Delta t}B\mathbf{U}^n+\mathbf{F}^n$, B fixed, $\mathbf{F}$ can change.

It's Lax-Richtmyer stable if $\|B^n\mathbf{w}\|\le C\|\mathbf{w}\|$ for all $\mathbf{w}\in\mathbb{R}^{N-1}$, $n\le\frac{T}{\Delta t}=N$

Property: If consistent with $\max_n\|\underline{\tau}^n\|_{2,\Delta x}=\mathcal{O}((\Delta x)^\beta)$ and L-R stable $\|B^n\mathbf{w}\|_{2,\Delta x}\le C\|\mathbf{w}\|_{2,\Delta x}$,

then, it convergents with $\max_n\|\mathbf{e}^n\|_{2,\Delta x}=\mathcal{O}((\Delta x)^\beta)$ Remark: **Always Assume $\Delta t=K(\Delta x)^\alpha$**

von Neumann Analysis: PDE: **1.** *periodic* BCs in spatial x, length D. **2.** 满足 L-R 中 PDE 框架, **3.** $\mathbf{F}^n=0$

$\Rightarrow$ Let $v_{k,m}=\exp(2\pi ik\frac{m\Delta x}{D})$ for $m=0,1,\dots,M-1$. Then vector $\mathbf{v}_k$ 为 B 的正交特征向量.

$\star$ By $U_m^n=\exp(2\pi ik\frac{m\Delta x}{D})$, $U_m^{n+1}=\alpha_kU_m^n$, 可得特征值(放大因子) α_k 表达式 (i.e. $B\mathbf{v}_k=\alpha_k\mathbf{v}_k$).

$\Rightarrow$ If all α_k for $k=0,1,\dots,M-1$ can be found $\Rightarrow B$ is normal and then:

If $|\alpha_k|\le1,\forall k\Rightarrow B$ is absolutely stable and L-R stable in weighted 2-norm

Remark: 假设 $\Delta t=K(\Delta x)^\alpha$ 会得 K,α 的限制条件, || For $\mathbf{F}^n\neq0$, 讨论 $\mathbf{F}^n=0$ 足够.

5 Boundary Condition

2nd ODE: For $u_xx=\beta(x)$ with $x\in(a,b)$, there are three types of BCs:

1. **Two Dirichlet:** $u(x=a)=c$ $u(x=b)=d$
2. **One Dirichlet, One Neumann:** $u(x=a)=c$ $u_x(x=b)=d$ or $u_x(x=a)=c$ $u(x=b)=d$
3. **One Neumann, One Constraint:** $u_x(x=a)=c$ $\int_a^b u(x)dx=d$

Heat PDE: For $u_t=\kappa u_{xx}$ with $x\in(a,b)$, there are three types of BCs: $\star$ **Basic IC:** $u(x,t_0)=P(x)$

1. **Two Dirichlet:** $u(x=a,t)=c$ $u(x=b,t)=d$
2. **One Dirichlet, One Neumann:** $u(x=a,t)=c$ $u_x(x=b,t)=d$ or $u_x(x=a,t)=c$ $u(x=b,t)=d$
3. **Two Neumann:** $u_x(x=a,t)=c$ $u_x(x=b,t)=d$

Remark: For $u_x(x=a,t)=c\Rightarrow$ descrite $\frac{u_1^n-u_0^n}{\Delta x}=c$ Boundary condition itself can effect the accuracy

For $u_t(x,t_0)=P(x)\Rightarrow$ descrite $\frac{u_m^1-u_m^0}{\Delta t}=P(a+m\Delta x)$

6 One Time Solve Two Variables PDEs

Elliptic Equation: $Lu=F(x_1,\dots,x_d)$ is an elliptic PDE if L is an elliptic operator.

Theorem: $L=\sum_{r=1}^d\sum s=1^dA_{r,s}\frac{\partial^2}{\partial x_r\partial x_s}+\sum_{n=1}^dB_n\frac{\partial}{\partial x_n}+C$ is elliptic operator if:

A is symmetric and A is positive|negative definite ($\lambda_i>0|\lambda_i<0$).

Poisson Equation: $\Delta u=F(x_1,\dots,x_d)$ is a Poisson equation and it's elliptic. with BCs:

Dirichlet: $u=u_D$ on $\partial\Omega$; Neumann: $\Delta u\cdot\hat{\mathbf{n}}=g$ on $\partial\Omega$ where $\int_\Omega F=\int_{\partial\Omega}g$ and $\int_\Omega u=\int_{\partial\Omega}u_D$.

e.g. For 2D: $u_{xx}+u_{yy}=F(x,y)$ with BCs: $u(x,y)=u_D$ on $\partial\Omega$

$\to$ **5-point Laplacian:** $\frac{U_{m-1,p}+U_{m+1,p}+U_{m,p-1}+U_{m,p+1}-4U_{m,p}}{(\Delta x)^2}=F(a+m\Delta x,a+p\Delta x)$

$\to$ Using $\mathbf{u}=[U_{1,1},U_{2,1},\dots,U_{M-1,1},U_{1,2},\dots,U_{M-1,M-1}]^\top$ and

$\mathbf{b}_{m+(p-1)(M-1)}=F(a+m\Delta x,a+p\Delta x)$ then $A\mathbf{u}=\mathbf{b}$ where A is a symmetric matrix.

$\to$ Eigenvalue/vectors of A: $\mathbf{v}_{k,l}=[v_{k,l,1,1},v_{k,l,2,1},\dots,v_{k,l,1,2},\dots,v_{k,l,M-1,M-1}]^\top$

$v_{k,l,m,p}=\sin(\pi km\Delta x)\sin(\pi lp\Delta x)$, $\lambda_{k,l}=\frac{2}{(\Delta x)^2}[\cos(\pi k\Delta x)+\cos(\pi l\Delta x)-2]<0$, $\lambda_{\min}=\lambda_{1,1}$

Remark: 这一节都是用同一个 Δx 离散 x 和 y 的, 和前面 t,x 分别不同.

Consistency: If $\lim_{\Delta x\to0}\|\underline{\tau}(\Delta x)\|=0$, the method is consistent.

Convergence: If $\|\mathbf{e}(\Delta x)\|=\mathcal{O}((\Delta x)^p)$, the method is p-th order accurate.

Stability: A numerical method is stable if for the linear system $A\mathbf{u}=\mathbf{b}$:

1. $A(\Delta x)$ is invertible $\forall\Delta x<\Delta x_0$
2. $\|[A(\Delta x)]^{-1}\mathbf{w}\|\le C\|\mathbf{w}\|$ for all $\mathbf{w}\in\mathbb{R}^{N-1}$

Property: If A normal, then $\|A^{-1}\mathbf{w}\|_2\le\frac{1}{|\lambda_1|}\|\mathbf{w}\|_2$ where $0<|\lambda_1|\le\cdots\le|\lambda_{N-1}|$ are eigenvalues of A.

7 Spectral Methods

Inner Product Space for Function: $\langle\phi_i,\phi_j\rangle=\int_a^b\phi_i\,\phi_j\,w\,dx$ ps: $w(x)$ is weight function.

For Fourier Series: $w(x)=1$, $a=-\pi$, $b=\pi$: $\sin(nx)$, $\cos(mx)$ are orthogonal basis.

Fourier Series: $f(x)=\frac{a_0}{2}+\sum_{n=1}^\infty a_n\cos(nx)+b_n\sin(nx)=\sum_{n=-\infty}^\infty A_ke^{ikx}$ where

$\to a_0=\frac1\pi\int_{-\pi}^\pi f(x)dx$ $a_n=\frac1\pi\int_{-\pi}^\pi f(x)\cos(nx)dx$ $b_n=\frac1\pi\int_{-\pi}^\pi f(x)\sin(nx)dx$.

$\to k>0$: $A_k=\frac12(a_k-ib_k)$; $k<0$: $A_k=\frac12(a_{-k}+ib_{-k})$; $k=0$: $A_0=\frac{a_0}{2}$. For real func: $A_{-k}=\overline{A_k}$

Property: $A_k=\frac{1}{2\pi}\int_{-\pi}^\pi f(x)e^{-ikx}dx$ and $f'(x)=\sum_{k=-\infty}^\infty(ik)A_ke^{ikx}$

More on Fourier Series: Assume f is periodic with period 2π, then:

1. The Fourier Transform: $\mathcal{F}\{f(x)\}=\widehat{f}(k)=\frac{1}{2\pi}\int_{-\pi}^\pi f(x)e^{-ikx}dx$
2. The inverse Fourier Transform: $\mathcal{F}^{-1}\{\widehat{f}(k)\}=f(x)=\sum_{k=-\infty}^\infty\widehat{f}(k)e^{ikx}$

Change $L\to2\pi$ period: If $f(x)$ has period $L\Rightarrow g(x)=f(\frac{L}{2\pi}x)$ has 2π period.

Discrete Fourier Transform (DFT): If $x_n=n\Delta x$ **equally-spaced** with $n=0,1,\dots,N-1$; $\Delta x=\frac{2\pi}{N}$:

1. **Fourier Transform:** $\mathcal{F}\{f\}=\widehat{f}_k=\frac{2\pi}{N}\sum_{n=0}^{N-1}f(x_n)e^{-ikx_n}$ $k=-\frac N2+1,\dots,\frac N2$
2. **Inverse FT:** $\mathcal{F}^{-1}\{\widehat{f}\}=f(x_n)=\frac{1}{2\pi}\sum_{k=-N/2+1}^{N/2}\widehat{f}_ke^{ikx_n}=\frac{1}{2\pi}\sum_{k=-N/2}^{N/2}{}'\widehat{f}_ke^{ikx_n}$

where ${}'\widehat{f}_k=\widehat{f}_k$ for $k=-\frac N2+1,\dots,\frac N2-1$; ${}'\widehat{f}_{-N/2}=\widehat{g}_{N/2}=\frac12\widehat{f}_{N/2}$.

Property: $\widehat{f'}_k=(ik)\widehat{f}_k\Leftrightarrow\mathcal{F}\{f'\}=(ik)\mathcal{F}\{f\}$

Remark: Quickly compute DFT: Fast Fourier Transform (FFT)

Using Fourier to solve PDE: For PDE with periodic BCs

1. Take Fourier Transform in PDE (in space) e.g. $u_t=u_{xx}\to\frac{d}{dt}\widehat{u}_k=-k^2\widehat{u}_k$
2. Take Fourier Transform in IC (in space) e.g. $u(x,0)=f(x)\to\widehat{u}_k(t=0)=\widehat{f}_k$
3. Using Transformed PDE and IC $\to$ a. Directly Solve ODE. b. numerical method for ODE. $\to\widehat{u}_k(t)$
4. Inverse Fourier Transform to get $u(x_n,t)$ 对 x_n 点 t 的估计或精确解.

Ideal of Interpolants: 给定 $(x_n,f(x_n))$, 构造插值函数 $p(x)$ s.t. $p(x_n)=f(x_n)\Rightarrow$ 用 $p'(x_n)$ 估计 $f'(x_n)$.

Spectral Collocation: Given $(x_n,f(x_n))$, f has period 2π, construct Fourier Interpolant $p(x)$ s.t. $p(x_n)=f(x_n)$

1. DFT: $\widehat{f}_k=\frac{2\pi}{N}\sum_{n=0}^{N-1}f(x_n)e^{-ikx_n}\Rightarrow p(x)=\frac{1}{2\pi}\sum_{k=-N/2}^{N/2}{}'\widehat{f}_ke^{ikx}$
2. (Method I) (即 FFT, 无法得到微分算子的矩阵) Direct differentiate $p(x)$, and $f'(x_n)\approx p'(x_n)$.
3. (Method II) Assume $p(x)=\sum_{m=0}^{N-1}f(x_m)P_\delta(x-x_m)$ where $P_\delta(x_j-x_m)=\delta_{j-m}$

$\Rightarrow P_\delta(x-x_m)=\frac{1}{2\pi}\sum_{k=-N/2}^{N/2}{}'\widetilde{\delta}_{k-m}e^{ikx}=\frac{1}{2\pi}\sum_{k=-N/2}^{N/2}{}'e^{ik(x-x_m)}$ and $P_\delta(x)=\frac{\sin(\pi x/\Delta x)}{(2\pi/\Delta x)\tan(x/2)}$

$\Rightarrow f'(x_n)\approx p'(x_n)=\sum_{m=0}^{N-1}f(x_m)P'_\delta(x_n-x_m)$ and $P'_\delta(x_n-x_m)=\begin{cases}0 & j\equiv0 \pmod N\\ \frac12(-1)^j\cot(\frac{jh}{2}) & j\not\equiv0 \pmod N\end{cases}$

$\Rightarrow$ Can be written in matrix form: $\mathbf{w}=D\mathbf{v}$ 其中 $\mathbf{w}$ 是 $f'(x_n)$ 的估计向量, v 是 $f(x_n)$ 的向量 and $f^{(n)}\Leftrightarrow D^n$

Chebyshev Spectral Method: For non-periodic BCs. Assume points are $x_n=\cos(\frac{n\pi}{N})$ for $n=0,1,\dots,N$.

Then $\mathbf{w}=D\mathbf{v}$ 其中 $\mathbf{w}$ 是 $f'(x_n)$ 的估计向量, v 是 $f(x_n)$ 的向量 and $f^{(n)}\Leftrightarrow D^n$ where $D_N\in\mathbb{R}^{(N+1)\times(N+1)}$ is:

$$D_N=\left[\begin{array}{c|c|c}\frac{2N^2+1}{6} & 2\frac{(-1)^j}{1-x_j} & \frac12(-1)^N\\ \hline -\frac12\frac{(-1)^i}{1-x_i} & \begin{matrix} & \frac{(-1)^{i+j}}{x_i-x_j}\\ -\frac{x_j}{2(1-x_j^2)} & \\ \frac{(-1)^{i+j}}{x_i-x_j} & \end{matrix} & \frac12\frac{(-1)^{N+i}}{1+x_i}\\ \hline -\frac12(-1)^N & -2\frac{(-1)^{N+j}}{1+x_j} & -\frac{2N^2+1}{6}\end{array}\right]$$

8 Iteration Method

JAC|GAU Method: Solve $Ax=b$ by iteration: $Px^{(k)}=(P-A)x^{(k-1)}+b$

1. **Jacobi:** $P=D$ **Gauss-Seidel:** $P=D+L$.
2. Error: $e^{(k)}=x^{(k)}-x$ and we have: $e^{(k)}=(I-P^{-1}A)^ke^{(0)}$
3. $e^{(k)}\to0$ as $k\to\infty\Leftrightarrow\|I-P^{-1}A\|<1\Leftrightarrow\rho(I-P^{-1}A)<1$ 收敛速度为 $-\log(\rho(I-P^{-1}A))$
4. By $\rho(A)\le\|A\|\Rightarrow\rho(A)\le\|A\|_\infty$, 可以由此估测一个收敛速度
5. Let ρ_J,ρ_{GS} 为 GAC|GAU 对应的 $\rho\Rightarrow$ **a.** If A **Tridigonal** (主对角和其上下两条对角线非零), then $\rho_{GS}=\rho_J^2$

   b. If *diagonal entries of A all positive and all other are non-positive* $\Rightarrow$ If $\rho_J<1$, then $0<\rho_{GS}<\rho_J<1$.

9 Appendix

x,x^2,x^3 **as** $x\to0$, x^{-1},x^{-2},x^{-3} **as** $x\to$ **inf**

9.1 Useful Tools

Taylor Series: $f(t+\Delta t)=f(t)+f'(t)\Delta t+\frac12f''(t)(\Delta t)^2+\cdots+\mathcal{O}((\Delta t)^N)$ **ps:** $u_{n+1}=u(n\Delta t+\Delta t)$

$u_m^{n+1}=u_m^n+\Delta t(u_t)_m^n+\frac{(\Delta t)^2}{2}(u_{tt})_m^n+\cdots$ where $(u_t)_m^n=\frac{\partial u}{\partial t}(a+m\Delta x,n\Delta t)$

$u_{m+1}^n=u_m^n+\Delta x(u_x)_m^n+\frac{(\Delta x)^2}{2}(u_{xx})_m^n+\cdots$ where $(u_x)_m^n=\frac{\partial u}{\partial x}(a+m\Delta x,n\Delta t)$

Useful Tools: If $e^n=Ae^{n-1}+B\tau^n$, then $e^n=A^ne^0+\sum_{k=1}^nA^{n-k}B\tau^k$, A: amplification factor.

Periodic BCs: Periodic boundary conditions for x: $u(a,t)=u(b,t)$, $u_x(a,t)=u_x(b,t)$ a,b 边界.

Heat Equation: For $u_t=\kappa u_{xx}$ with two Dirichlet BCs

	Forard Euler	Backward Euler	Crank-Nicolson
Solve Matrix?	No	Yes	Yes
r_m^n	$\mathcal{O}(\Delta t)+\mathcal{O}((\Delta x)^2)$	$\mathcal{O}(\Delta t)+\mathcal{O}((\Delta x)^2)$	$\mathcal{O}((\Delta t)^2)+\mathcal{O}((\Delta x)^2)$
Stability Condition	$\kappa\frac{\Delta t}{(\Delta x)^2}\le\frac12$	None	None

9.2 NLA

Matrix Norm: $\|A\|=\max_{\|\mathbf{x}\|=1}\|A\mathbf{x}\|$ $\|AB\|\le\|A\|\|B\|$ $\|A+B\|\le\|A\|+\|B\|$ $\|\lambda A\|=|\lambda|\|A\|$

$\|A\|_\infty=\max_{1\le i\le m}\sum_{j=1}^n|a_{ij}|$ (最大行和) $\|A\|_1=\max_{1\le j\le n}\sum_{i=1}^m|a_{ij}|$ (最大列和)

$\|A\|_2=\sqrt{\rho(A^TA)}$ where $\rho(A)=\max_i|\lambda_i(A)|$ $\rho(A)\le\|A\|$ $\|A^k\|\to0$ **as** $k\to\infty\Leftrightarrow\rho(A)<1$

10 The Steepest Descent Method

Theorem: Let $\phi(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T A\mathbf{x} - \mathbf{b}^T\mathbf{x}$, then it has unique minimum if A is *positive definite*.

If A is symmetric positive definite $\Rightarrow$ $\mathbf{x}$: $A\mathbf{x} = \mathbf{b} \Leftrightarrow \mathbf{x}$: $\phi(\mathbf{x}) = \min_{\tilde{\mathbf{x}}} \phi(\tilde{\mathbf{x}}) = \min_{\tilde{\mathbf{x}}}[\frac{1}{2}\tilde{\mathbf{x}}^T A\tilde{\mathbf{x}} - \mathbf{b}^T\tilde{\mathbf{x}}]$

Following need A to be symmetric positive definite.

Steepest Descent Method: To solve $\min \phi(x)$ or $A\mathbf{x} = \mathbf{b}$, we can use the following iteration:

$\mathbf{x}^{(k)} = \mathbf{x}^{(k-1)} - \alpha_{k-1}\nabla\phi(\mathbf{x}^{(k-1)})$ and α_{k-1} s.t. $\min_\alpha \phi(\mathbf{x}^{(k-1)} - \alpha\nabla\phi(\mathbf{x}^{(k-1)}))$.

1. **Property 1**: $\nabla\phi(\mathbf{x})$ is the direction of steepest ascent (most rapid increase) of ϕ at $\mathbf{x}$.
2. **Property 2**: $\alpha_{k-1} = \frac{(\mathbf{r}^{(k-1)})^\top \mathbf{r}^{(k-1)}}{(\mathbf{r}^{(k-1)})^\top A\mathbf{r}^{(k-1)}}$ and $\nabla\phi(\mathbf{x}^{(k)}) = A\mathbf{x}^{(k)} - \mathbf{b} =: -\mathbf{r}^{(k)}$ is **k-th residual**.
3. **Property 3**: $\mathbf{r}^{(k)} = \mathbf{b} - A\mathbf{x}^k = \cdots = \mathbf{r}^{(k-1)} - \alpha_{k-1}A\mathbf{r}^{(k-1)}$

Remark: Convergence behaviour is *highly problem-dependent*. (如果 ϕ 的等高线是圆形则快, 椭圆形则慢)

The Conjugate Gradient Method: To solve $\min \phi(x)$ or $A\mathbf{x} = \mathbf{b}$, we can use the following iteration:

$\mathbf{x}^{(k)} = \mathbf{x}^{(k-1)} + \alpha_{k-1}\mathbf{p}^{(k-1)}$ and α_{k-1} s.t. $\min_\alpha \phi(\mathbf{x}^{(k-1)} + \alpha\mathbf{p}^{(k-1)})$

and $\mathbf{p}^{(k)}$ is *A-conjugate* to previous **search direction**: $\mathbf{p}^{(k-1)}$: $(\mathbf{p}^{(k)})^\top A\mathbf{p}^{(k-1)} = 0$

1. **Property 1**: $\alpha_{k-1} = \frac{(\mathbf{r}^{(k-1)})^\top \mathbf{r}^{(k-1)}}{(\mathbf{p}^{(k-1)})^\top A\mathbf{p}^{(k-1)}}$ and $-\mathbf{r}^{(k)} = A\mathbf{x}^{(k)} - \mathbf{b}$ is **k-th residual**.
2. **Property 2**: $\mathbf{p}^{(k)} = \mathbf{r}^{(k)} + \beta_{k-1}\mathbf{p}^{(k-1)}$ where $\beta_{k-1} = \frac{(\mathbf{r}^{(k)})^\top \mathbf{r}^{(k)}}{(\mathbf{r}^{(k-1)})^\top \mathbf{r}^{(k-1)}}$
3. **Property 3**: $(\mathbf{p}^{(k)})^\top A\mathbf{p}^{(j)} = 0$ and $(\mathbf{r}^{(k)})^\top \mathbf{r}^{(j)} = 0$ for $j = 0, 1, \ldots, k-1$
4. **Property 4**: $\text{span}\{\mathbf{p}^{(0)}, \ldots, \mathbf{p}^{(k-1)}\} = \text{span}\{A^0\mathbf{r}^{(0)}, \ldots, A^{k-1}\mathbf{r}^{(0)}\} = \text{span}\{A\mathbf{e}^{(0)}, \ldots, A^k\mathbf{e}^{(0)}\}$

   where $\mathbf{e}^{(k)} = \mathbf{x}^{(k)} - \mathbf{x}$ is the error at k-th iteration. (Quicker than JAC|GAU)
5. **Exact Property**: For $A \in \mathbb{R}$ has n *distinct* eigenvalues $\Rightarrow$ solve in **at most n** iterations. (By exact arithmetic)

A-Norm: The A-norm of $\mathbf{x}$ is defined as $\|\mathbf{x}\|_A = \sqrt{\mathbf{x}^T A\mathbf{x}}$.

Condition Number: The condition number of A is $\kappa(A) = \|A\|\|A^{-1}\| \geq 1$ and $\kappa_2(A) = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)}$

Error of Conjugate Gradient: To solve $A\mathbf{x} = \mathbf{b}$, error is: $\frac{\|\mathbf{e}^{(k)}\|_A}{\|\mathbf{e}^{(0)}\|_A} \leq 2\left(\frac{\sqrt{\kappa(A)}-1}{\sqrt{\kappa(A)}+1}\right)^k$

11 The advection equation

Hyperbolic system: For $\mathbf{u}_t + A\mathbf{u}_x = 0$, if A is diagonalizable with real eigenvalues $\Rightarrow$ hyperbolic system.

Diagonalization: If $P^{-1}AP = \Lambda$, then Let $\mathbf{v} = P^{-1}\mathbf{u}$, we have $\mathbf{v}_t + \Lambda\mathbf{v}_x = 0$

Advection Equation: The advection equation is $u_t + vu_x = 0$ where v is the advection speed.

Translate to Wave Equation: 两边对 t 求导后代入原式可得 $u_{tt} = v^2 u_{xx}$

- **If IC is $u(x,0) = P(x)$**: solution is $u(x,t) = P(x - vt)$
  - Moreover, **if BC is Periodic**: sol $u(x,t) = \widetilde{P}(x-vt)$ where $\widetilde{P}(x+k(b-a)) = P(x)$ for $x \in [a,b]$, $k \in \mathbb{Z}$.
- **If BC is $u(a,t) = P(t)$**: solution is $u(x,t) = P(t - \frac{x-a}{v})$

Remark: Periodic BCs $\Rightarrow U_p = U_{M+p}$ for $p = -1, 0$

Courant Number: The Courant number is defined as $C = \frac{v\Delta t}{\Delta x}$.

Characteristics Based Method: For the advection equation $u_t + vu_x = 0$, and $|C| = |\frac{v\Delta t}{\Delta x}| \leq 1$:

$\Rightarrow u(x_m, t_{n+1}) = u(x_m - v\Delta t, t_n) \approx U_m^n + \frac{|v|+v}{2}\frac{\Delta t}{\Delta x}(U_{m-1}^n - U_m^n) + \frac{|v|-v}{2}\frac{\Delta t}{\Delta x}(U_{m+1}^n - U_m^n) =: U_m^{n+1}$

Equivalently, $\frac{U_m^{n+1}-U_m^n}{\Delta t} + \frac{|v|+v}{2}\frac{U_m^n-U_{m-1}^n}{\Delta x} - \frac{|v|-v}{2}\frac{U_{m+1}^n-U_m^n}{\Delta x} = 0$

Remark: Above is for linear interpolation. (e.g. For $v > 0$, $u(x_m, t_{n+1}) \approx U_m^{n+1} = (1-C)U_m^n + CU_{m-1}^n$)

Upwinding: For the advection equation $u_t + vu_x = 0$:

For $v > 0$, use $U_m^n, U_{m-1}^n, U_{m-2}^n, \ldots$ to estimate $u(x_m - v\Delta t, t_n) = u(x_m, t_{n+1})$, denote by U_m^{n+1}.

For $v < 0$, use $U_m^n, U_{m+1}^n, U_{m+2}^n, \ldots$ to estimate $u(x_m - v\Delta t, t_n) = u(x_m, t_{n+1})$, denote by U_m^{n+1}.

$\Rightarrow$ Thus $v > 0 \to \frac{dU_m}{dt} + \frac{v}{\Delta x}\sum_{q=0}^Q \gamma_q U_{m-q} = 0$ (i.e. $\frac{du}{dx}|_{x_m} \approx \frac{1}{\Delta x}\sum_{q=0}^Q \gamma_q U_{m-q}$)

$\Rightarrow$ Thus $v < 0 \to \frac{dU_m}{dt} + \frac{v}{\Delta x}\sum_{q=0}^Q \gamma_q U_{m+q} = 0$ (i.e. $\frac{du}{dx}|_{x_m} \approx \frac{1}{\Delta x}\sum_{q=0}^Q \gamma_q U_{m+q}$)

ps: For $u_t + vu_x = 0 \to$ In characteristics form: $\frac{du}{dt} = 0$ on $\frac{dx}{dt} = v \Rightarrow u = A$ on $x = vt + B$

12 Appendix

12.1 Other

Divergence Theorem: $\int_\Omega \nabla\cdot\mathbf{F}dV = \int_{\partial\Omega}\mathbf{F}\cdot\hat{\mathbf{n}}dS$ $\hat{\mathbf{n}}$ outward normal. $\nabla^2 = \nabla\cdot\nabla = \Delta$

How to improve Approximation Accuracy?: a. Using more neighboring points. (more terms from Taylor expansion) b. Make Δt, Δx smaller. c. For Periodic BCs, use Fourier Transform.

12.2 ODE

$y' + Py = Q$: $\mu y = \int Q(x)\mu dx + C$, $\mu(x) = e^{\int P(x)dx}$ $y' = Py$: $y = Ce^{\int P(x)dx}$

$y'' + k^2 y = 0$: $y = A\cos(kx) + B\sin(kx)$ $y'' - k^2 y = 0$: $y = Ae^{kx} + Be^{-kx}, k > 0$

$ay'' + by' + cy = 0$: Characteristic equation: $ar^2 + br + c = 0$ $r_{1,2} = \mu \pm i\nu$

a. $\Delta > 0$: $y = Ae^{r_1 x} + Be^{r_2 x}$; **b.** $\Delta = 0$: $y = (A + Bx)e^{\mu x}$; **c.** $\Delta < 0$: $y = e^{\mu x}(A\cos(\nu x) + B\sin(\nu x))$

12.3 Integral

Gaussian Integral: $\int_{-\infty}^{\infty} e^{-ax^2}dx = \frac{\sqrt{\pi}}{\sqrt{a}}$ $\int_0^\infty e^{-xt^n}t^m = \frac{1}{n}\Gamma(\frac{m+1}{n})x^{-\frac{m+1}{n}}$

Integral: $\int\tan x = -\ln|\cos x| = \ln|\sec x|$ $\int\cot x = \ln|\sin x| = -\ln|\csc x|$

Techniques: **a.** e.g. $\int e^{at}\cos(bt)dt = \text{Re}(\int e^{(a+ib)t}dt)$

Basis: For $\langle\cdot,\cdot\rangle$ with $\langle f, g\rangle = \int_a^b f(x)g(x)dx$ ($\rho = 1$) we have: $\langle\sin(n\pi(x-a)/L), \cos(m\pi(x-b)/L)\rangle = 0$

For $\langle\sin(n\pi(x-a)/L), \sin(m\pi(x-b)/L)\rangle = \frac{L}{2}\delta_{nm}$, $\langle\cos(n\pi(x-a)/L), \cos(m\pi(x-b)/L)\rangle = \frac{L}{2}\delta_{nm}$.

12.4 Expansion

Binomial Expansion: $(1+x)^\alpha \sim 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \cdots$ as $x \to 0$

For $x \to a$: Taylor at $x = a$: $f(x) \sim f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots$ as $x \to a$

Taylor|$x \to 0$: $f(x+h) \sim f(x) + f'(x)h + \frac{f''(x)}{2!}h^2 + \cdots$ as $h \to 0$

$\sin(x) \sim x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$ as $x \to 0$ $e^x \sim 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$ as $x \to 0$

$\cos(x) \sim 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$ as $x \to 0$ $\ln(1+x) \sim x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$ as $x \to 0$

$\sinh(x) \sim x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots$ as $x \to 0$ $\cosh(x) \sim 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots$ as $x \to 0$

$\tan(x) \sim x + \frac{x^3}{3} + \frac{2x^5}{15} + \cdots$ as $x \to 0$ $\cot(x) \sim \frac{1}{x} - \frac{x}{3} - \frac{x^3}{45} - \cdots$ as $x \to 0$

For $x \to \infty$: $\arctan(x) \sim \frac{\pi}{2} - \frac{1}{x} + \frac{1}{3x^3} - \cdots$ as $x \to \infty$

$\ln(a+x) = \ln x + \ln(1 + \frac{a}{x}) \sim \ln x + \frac{a}{x} - \frac{a^2}{2x^2} + \cdots$ as $x \to \infty$

$\frac{1}{a+x} = \frac{1}{x}\cdot\frac{1}{1+\frac{a}{x}} \sim \frac{1}{x} - \frac{a}{x^2} + \frac{a^2}{x^3} - \cdots$ as $x \to \infty$

sin|cos: $a\sin(x) + b\cos(x) = \sqrt{a^2+b^2}\sin(x+\varphi)$ $a\cos(x) + b\sin(x) = \sqrt{a^2+b^2}\cos(x-\varphi)$, φ 过 (a, b).

- $\sin(x) + \sin(y) = 2\sin(\frac{x+y}{2})\cos(\frac{x-y}{2})$ | $\sin(x)\cos(y) = \frac{1}{2}[\sin(x+y) + \sin(x-y)]$ s+s=sc,s-s=cs,c+c=cc,c-c=-ss

Inequalities: $\ln(x+1) < x$, $e^x > x+1$, $(1+x)^a < e^{ax}$

12.5 Other

Nabla: $\nabla^2 u = u_{xx} + u_{yy} + u_{zz} + \cdots$ $\nabla\cdot\mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$

Binomial Expansion: $(a+b)^n = \sum_{k=0}^n \binom{n}{k}a^{n-k}b^k$ $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ $(1+x)^n = \sum_{k=0}^n\binom{n}{k}x^k$

Sin/Cos: $\sin[\frac{2n+1}{2}\pi] = (-1)^n$ $\cos(n\pi) = (-1)^n$

13 Some Proofs

§10 Theorem Proof: $\phi(x) = \min_{\widetilde{x}}\phi(\widetilde{x}) = \min_{\widetilde{x}}[\frac{1}{2}\widetilde{x}^T A\widetilde{x} - \mathbf{b}^T\widetilde{x}]$, A symmetric positive definite, x solves $Ax = b$.

($\Rightarrow$) Let x solve $Ax = b$, then $\phi(x) = \frac{1}{2}\underbrace{\left(A^{\frac{1}{2}}x - A^{-\frac{1}{2}}b\right)^\top\left(A^{\frac{1}{2}}x - A^{-\frac{1}{2}}b\right)}_{=\|A^{1/2}x - A^{-1/2}b\|_2^2} - \underbrace{\frac{1}{2}b^T A^{-1}b}_{\text{constant}}$

As $Ax = b$, then $A^{\frac{1}{2}}x - A^{-\frac{1}{2}}b = 0$, thus $\phi(x) = -\frac{1}{2}b^T A^{-1}b$ achieves the minimum.

($\Leftarrow$) Let $\phi(x) = \min_{\widetilde{x}}\phi(\widetilde{x})$, then $\nabla\phi(x) = Ax - b = 0$, thus $Ax = b$.

§10 Why choose α_{k-1} s.t. $\min_\alpha \phi(\mathbf{x}^{(k-1)} + \alpha\mathbf{r}^{(k-1)})$: Steepest descent method

Since $\phi(\mathbf{x}^{(k-1)} + \alpha\mathbf{r}^{(k-1)}) = [\frac{1}{2}(\mathbf{x}^{(k-1)})^\top A\mathbf{x}^{(k-1)} - \mathbf{b}^\top\mathbf{x}^{(k-1)}]$
$+\alpha[\mathbf{x}^{(k-1)\top}A\mathbf{r}^{(k-1)} - \mathbf{b}^\top\mathbf{r}^{(k-1)}] + \frac{\alpha^2}{2}(\mathbf{r}^{(k-1)})^\top A\mathbf{r}^{(k-1)}$

So that $\frac{d\phi}{d\alpha} = (\mathbf{x}^{(k-1)})^\top A\mathbf{r}^{(k-1)} - \mathbf{b}^\top\mathbf{r}^{(k-1)} + \alpha(\mathbf{r}^{(k-1)})^\top A\mathbf{r}^{(k-1)}$
$= (A\mathbf{x}^{(k-1)} - \mathbf{b})^\top\mathbf{r}^{(k-1)} + \alpha(\mathbf{r}^{(k-1)})^\top A\mathbf{r}^{(k-1)} = -(\mathbf{r}^{(k-1)})^\top\mathbf{r}^{(k-1)} + \alpha(\mathbf{r}^{(k-1)})^\top A\mathbf{r}^{(k-1)}$

Set $\frac{d\phi}{d\alpha} = 0$, we have $\alpha_{k-1} = \frac{(\mathbf{r}^{(k-1)})^\top\mathbf{r}^{(k-1)}}{(\mathbf{r}^{(k-1)})^\top A\mathbf{r}^{(k-1)}}$

§10 Why choose α_{k-1} s.t. $\min_\alpha \phi(\mathbf{x}^{(k-1)} + \alpha\mathbf{p}^{(k-1)})$: Conjugate gradient method

Since $\phi(\mathbf{x}^{(k-1)} + \alpha\mathbf{p}^{(k-1)}) = [\frac{1}{2}(\mathbf{x}^{(k-1)})^\top A\mathbf{x}^{(k-1)} - \mathbf{b}^\top\mathbf{x}^{(k-1)}]$
$+\alpha[\mathbf{x}^{(k-1)\top}A\mathbf{p}^{(k-1)} - \mathbf{b}^\top\mathbf{p}^{(k-1)}] + \frac{\alpha^2}{2}(\mathbf{p}^{(k-1)})^\top A\mathbf{p}^{(k-1)}$

So that $\frac{d\phi}{d\alpha} = (\mathbf{x}^{(k-1)})^\top A\mathbf{p}^{(k-1)} - \mathbf{b}^\top\mathbf{p}^{(k-1)} + \alpha(\mathbf{p}^{(k-1)})^\top A\mathbf{p}^{(k-1)}$
$= (A\mathbf{x}^{(k-1)} - \mathbf{b})^\top\mathbf{p}^{(k-1)} + \alpha(\mathbf{p}^{(k-1)})^\top A\mathbf{p}^{(k-1)} = -(\mathbf{r}^{(k-1)})^\top\mathbf{p}^{(k-1)} + \alpha(\mathbf{p}^{(k-1)})^\top A\mathbf{p}^{(k-1)}$

Set $\frac{d\phi}{d\alpha} = 0$, we have $\alpha_{k-1} = \frac{(\mathbf{r}^{(k-1)})^\top\mathbf{p}^{(k-1)}}{(\mathbf{p}^{(k-1)})^\top A\mathbf{p}^{(k-1)}}$ As $(\mathbf{r}^{(k-1)})^\top\mathbf{p}^{(k-1)} = (\mathbf{r}^{(k-1)})^\top\mathbf{r}^{(k-1)}$ so we have $\alpha_{k-1} = \ldots$

Show that: $(\mathbf{r}^{(k-1)})^\top\mathbf{p}^{(k-1)} = (\mathbf{r}^{(k-1)})^\top\mathbf{r}^{(k-1)}$.

Since $\mathbf{p}^{(k)} = \mathbf{r}^{(k)} + \beta_{k-1}\mathbf{p}^{(k-1)}$, we have $(\mathbf{r}^{(k-1)})^\top\mathbf{p}^{(k-1)} = (\mathbf{r}^{(k-1)})^\top\mathbf{r}^{(k-1)} + \beta_{k-2}(\mathbf{r}^{(k-1)})^\top\mathbf{p}^{(k-2)}$.

As $(\mathbf{r}^{(k-1)})^\top\mathbf{p}^{(k-2)} = 0$, we have $(\mathbf{r}^{(k-1)})^\top\mathbf{p}^{(k-1)} = (\mathbf{r}^{(k-1)})^\top\mathbf{r}^{(k-1)}$.

Show that: $(\mathbf{r}^{(k-1)})^\top\mathbf{p}^{(k-2)} = 0$.

Since $\phi(\alpha_{k-2}) = \frac{1}{2}(\mathbf{x}^{(k-2)} + \alpha_{k-2}\mathbf{p}^{(k-2)})^\top A(\mathbf{x}^{(k-2)} + \alpha_{k-2}\mathbf{p}^{(k-2)}) - \mathbf{b}^\top(\mathbf{x}^{(k-2)} + \alpha_{k-2}\mathbf{p}^{(k-2)})$.

By using $A = A^\top$, we have $\phi'(\alpha_{k-2}) = (\mathbf{x}^{(k-2)})^\top A\mathbf{p}^{(k-2)} - \mathbf{b}^\top\mathbf{p}^{(k-2)} + \alpha_{k-2}(\mathbf{p}^{(k-2)})^\top A\mathbf{p}^{(k-2)}$
$= (A\mathbf{x}^{(k-2)} - \mathbf{b})^\top\mathbf{p}^{(k-2)} + \alpha_{k-2}(\mathbf{p}^{(k-2)})^\top A\mathbf{p}^{(k-2)}$

Since $\phi'(\alpha_{k-2}) = 0$, we have $(A\mathbf{x}^{(k-2)} - \mathbf{b})^\top\mathbf{p}^{(k-2)} + \alpha_{k-2}(\mathbf{p}^{(k-2)})^\top A\mathbf{p}^{(k-2)} = 0$.

So that $(A(\mathbf{x}^{(k-2)} + \alpha_{k-2}\mathbf{p}^{(k-2)}) - \mathbf{b})^\top\mathbf{p}^{(k-2)} = 0$.

As $\mathbf{x}^{(k-1)} = \mathbf{x}^{(k-2)} + \alpha_{k-2}\mathbf{p}^{(k-2)}$ and $-\mathbf{r}^{(k-1)} = A\mathbf{x}^{(k-1)} - \mathbf{b}$, we have $(\mathbf{r}^{(k-1)})^\top\mathbf{p}^{(k-2)} = 0$.

§1.2 Norm: For $d = 1$, $\Delta x = \frac{D}{N}$, $u_i = \mathcal{O}((\Delta x)^p)$, then $\|\mathbf{u}\|_{2,\Delta x} = \mathcal{O}((\Delta x)^p)$.

Proof: $\|\mathbf{u}\|_{2,\Delta x}^2 = \Delta x\sum_{i=1}^{N-1}u_i^2 \leq \Delta x(N-1)\max_{1\leq i\leq N-1}u_i^2 \leq D\max_{1\leq i\leq N-1}u_i^2$, where $\Delta x = \frac{D}{N}$.

Thus $\|\mathbf{u}\|_{2,\Delta x} = \mathcal{O}((\Delta x)^p)$.

Proof: If $u = [U_{1,1}, U_{2,1}, \ldots, U_{1,2}, U_{2,2}, \ldots, U_{M-1,M-1}]^\top$, $d = 2$ (x,y), $x = m\Delta x$, $y = p\Delta x$,

$\|\mathbf{u}\|_{2,\Delta x}^2 = (\Delta x)^2\sum_{m=1}^{M-1}\sum_{p=1}^{M-1}U_{m,p}^2 \leq (\Delta x)^2(M-1)^2\max_{1\leq m,p\leq M-1}U_{m,p}^2$
$\leq D^2\max_{1\leq i,j\leq M-1}U_{i,j}^2$, where $\Delta x = \frac{D}{M}$.

Thus $\|\mathbf{u}\|_{2,\Delta x} = \mathcal{O}((\Delta x)^p)$.